

What Impact Does Gasoline Taxes have on Prices and Sales of Gasoline?

DSC 630 Predictive Analytics

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Gasoline Tax Impact on Price and Sales

- Country relies heavily on gasoline
- 17% of the retail price of gasoline is taxes
- How do taxes impact the price of gasoline?

Data Selection

Using publicly available data sets:

- Tax Rates by Motor Fuel and State
 - 11 columns with annual data for 2,332 rows
- Monthly Motor Fuel Sales Reported by States
 - 8 columns with 15,252 rows
- Weekly Retail Gasoline and Diesel Prices
 - 4 data sets based on geolocation
- Gross Domestic Product
 - Monthly data from 1997 to current
- Data then complied to sync up data elements with time frames.

Seq	state	Gallons_Sold	Date	Retail Prices	Fed_Tax_Rate	CA_Tax_Rate	GEPUCURRENT
97	California	1,224,282,334	20-Jan	3.489	0.184	0.473	194.4258
98	California	1,197,585,656	20-Feb	3.447	0.184	0.473	198.7127
99	California	1,019,045,856	20-Mar	3.262	0.184	0.473	320.3234
100	California	714,431,599	20-Apr	2.827	0.184	0.473	331.6438
101	California	924,817,479	20-May	2.771	0.184	0.473	420.8612
102	California	1,029,938,226	20-Jun	2.972	0.184	0.473	314.3776
103	California	1,089,575,995	20-Jul	3.102	0.184	0.505	317.0378
104	California	1,120,836,492	20-Aug	3.156	0.184	0.505	294.3455
105	California	999,449,673	20-Sep	3.174	0.184	0.505	314.9368
106	California	1,123,313,104	20-Oct	3.137	0.184	0.505	310.2022
107	California	1,039,176,930	20-Nov	3.115	0.184	0.505	405.0704
108	California	1,029,256,230	20-Dec	3.143	0.184	0.505	307.4836
109	California	995,980,173	21-Jan	3.262	0.184	0.505	276.5787
110	California	975,834,460	21-Feb	3.434	0.184	0.505	197.5037
111	California	1,138,915,242	21-Mar	3.781	0.184	0.505	215.9586
112	California	1,155,077,720	21-Apr	3.911	0.184	0.505	198.3213
113	California	1,209,075,052	21-May	4.068	0.184	0.505	176.2512
114	California	1,196,780,811	21-Jun	4.182	0.184	0.505	173.7252
115	California	1,232,804,307	21-Jul	4.253	0.184	0.511	192.1818
116	California	1,235,150,159	21-Aug	4.319	0.184	0.511	205.5088
117	California	1,186,156,014	21-Sep	4.31	0.184	0.511	215.7342
118	California	1,143,477,704	21-Oct	4.4	0.184	0.511	191.0006
119	California	1,163,528,519	21-Nov	4.598	0.184	0.511	233.5518
120	California	1,203,056,833	21-Dec	4.597	0.184	0.511	263.9804

Risks

- Data not showing a correlation, state specific regulations, and economic conditions.

Ethical Considerations

- For ethical implications, taxes can be a controversial topic, especially in the political world, analysis must be done so with no political bias.

Contingency Plan

- Find different data sets:
 - crude oil production
 - socio-economic factors such as
 - Unemployment
 - consumer confidence
 - consumer spending
 - inflation
 - currency rates
 - population changes
 - political changes
- Adjust my problem statement

Risks Ethical Considerations Contingency Plan

Data Models and Evaluation Plan

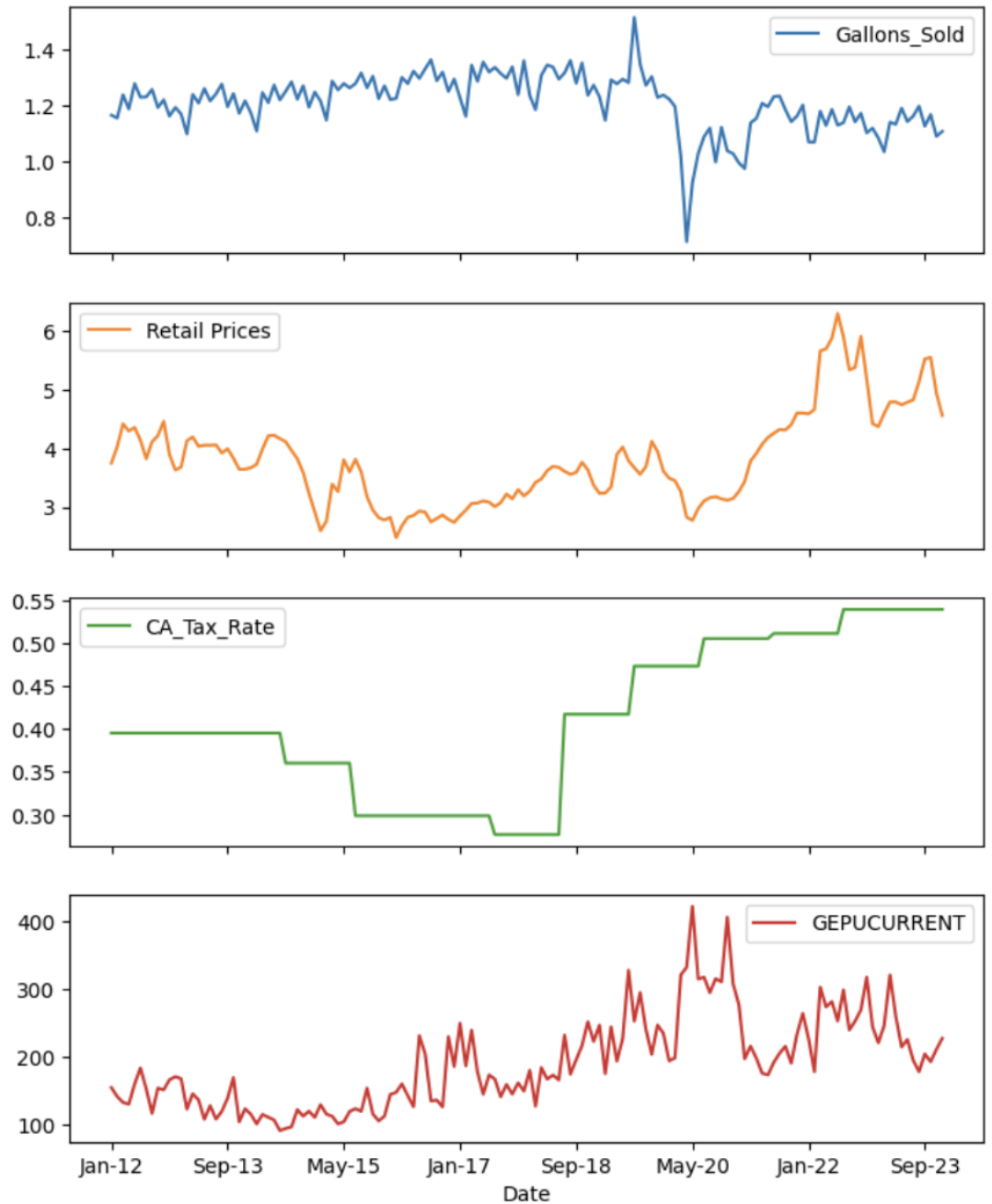
Model Type

- Linear Regression
- Ridge Regression
- Lasso Regression
- Random Forest

Evaluation Metrics

- Root Mean Square Error (RMSE)
- R-squared (R^2)

California Data

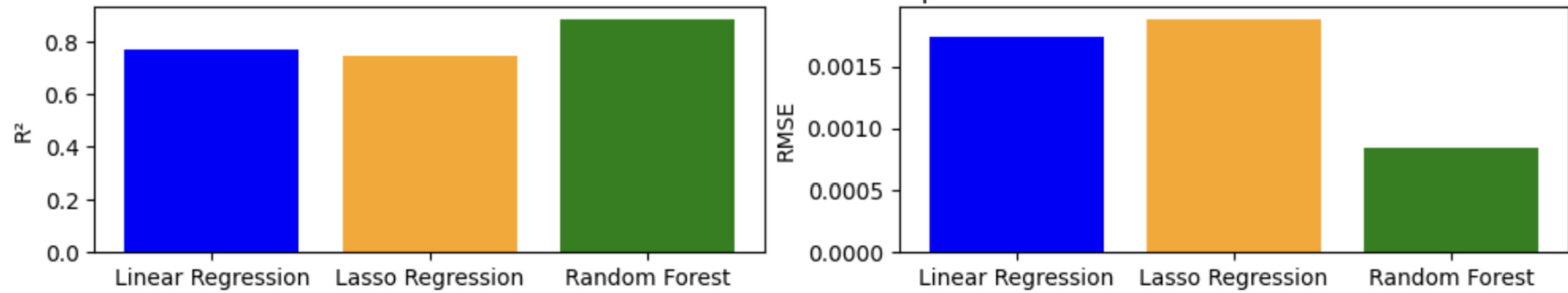


Confusion Matrix

The California tax rate has:

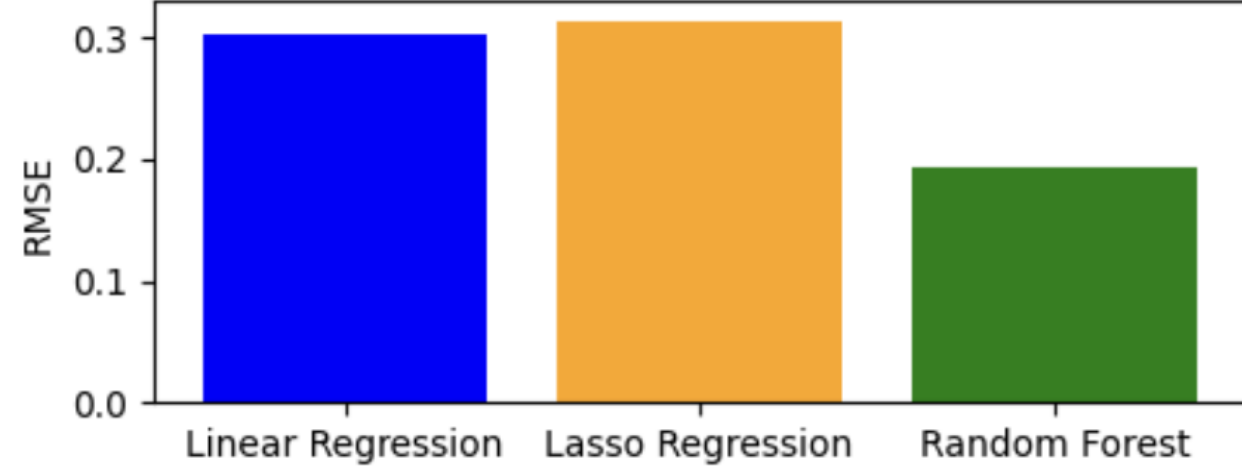
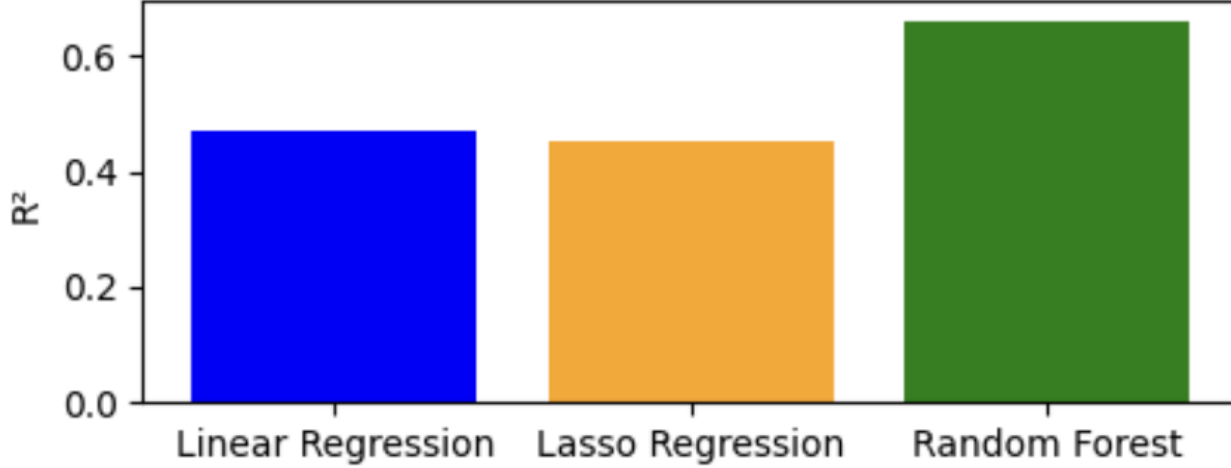
- a negative impact on the number of gallons sold
- a positive impact on the retail price
- a positive relationship to the GDP.

	Gallons_Sold	Retail Prices	CA_Tax_Rate	GDP
Gallons_Sold	1.000000	-0.219050	-0.593143	-0.479213
Retail Prices	-0.219050	1.000000	0.666901	0.186847
CA_Tax_Rate	-0.593143	0.666901	1.000000	0.597674
GDP	-0.479213	0.186847	0.597674	1.000000



Model Performance

I started my analysis looking at the impact of price, gallons sold, and GDP influencing on predicting tax rate. These graphs show the difference in the model R^2 and RMSE results. When the R^2 is closer to 1 the model is a better fit to the data. The desired RMSE is closer to zero.



2nd Model Performance

After looking at the results of these, I decided to go back and rerun the same models but where prices were the target and only taxes is the only feature. This would better answer the question: “What impact does gasoline taxes have on prices and sales of gasoline”.

Model Interpretation

First regression model had comparable results in R^2 were about .75 and low RSME indicating high fit and little variance in the model. The Random Forest mode had an R^2 of .89 with an RSME 50% lower than the regression models. But this model had tax rate as the target variable.

The second model was run with the target variable of price and the tax amount as the only other model variable. This time the R^2 were much lower at about 50% of the first model with higher variability in the RSME.

Conclusion

- There are relationships between taxes and the other features
- Using the retail price, gallons sold, and GDP can predict the tax rate
- The tax rate alone and its ability to predict the retail price does not have strong models.
- My initial proposal was that as taxes go up so should the retail price leading to decreased sales. The models have shown that they could predict the retail price based on taxes but with much variance. This tells me that the other features are needed to accurately predict retail prices and volume of gasoline sold.
- To be able to deploy a model for this topic, additional models would need to be run.



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